Cosmic Queries – Curiosity, Conspiracies & the God Particle with Jordan Klepper

Podcast: Startalk Podcast

AI Executive Summary

This segment of StarTalk Cosmic Queries with Jordan Klepper featured a lively discussion between Neil deGrasse Tyson, a renowned astrophysicist, and Jordan Klepper, a comedian and political commentator. The conversation covered a wide range of topics, including the importance of arts education, the future of education, the impact of confirmation bias, and the effects of space debris on the night sky.

Key Points:

1. **Arts Education and Its Value:**
2. Neil deGrasse Tyson emphasized the importance of arts education, arguing that it enriches one's life and provides a different perspective on the world.

3. Jordan Klepper agreed, noting that the arts can help articulate feelings and emotions that are difficult to express otherwise.

4. **Confirmation Bias:**

5. Both agreed that confirmation bias is a significant issue in today's society, where people often only seek out information that supports their existing beliefs.

6. Neil suggested that teaching a class on psychological biases in school could help mitigate this problem.

7. **Space Debris and the Night Sky:**

8. Neil discussed the impact of space debris, particularly the SpaceX Starlink satellites, on the night sky and the loss of the "sacred" night sky.

9. He expressed concern about the future of the night sky and the potential loss of the cosmic perspective it provides.

10. **Vaccines and Anti-Vax Movements:**

11. Jordan addressed the issue of vaccine hesitancy, recommending a documentary called "Shot in the Arm" to explore the reasons behind vaccine hesitancy and how to overcome it.

12. **Performing and Public Speaking:**

13. Jordan shared his experiences with performing and public speaking, noting that the fear of failure and the need to connect with the audience are integral parts of the process.

14. Neil provided insights into how scientists and educators handle failure and the importance of maintaining a scientific mindset.

15. **Space and the Cosmos:**

16. The conversation touched on various space-related topics, including the impact of a hypothetical displacement of the solar system and the effects of a sudden disappearance of the sun.

Overall Impression:

The discussion was engaging and thought-provoking, blending scientific insights with personal anecdotes and political commentary. Both Neil and Jordan demonstrated a deep understanding of their respective fields and a commitment to fostering critical thinking and rational discourse.

Final Thoughts:

The segment highlighted the importance of interdisciplinary approaches to education and the value of maintaining a scientific and open-minded perspective in a rapidly changing world. It also underscored the role of entertainment and comedy in addressing complex issues and engaging the public in meaningful conversations.

Diarized Transcript

SPEAKER_00: You know that idea you can't stop thinking about. The one you want to turn into a business. It's time. Shopify has everything you need to launch your website fast and sell everywhere customers shop. We just made it live and it's already sold out. Join millions of Shopify merchants who have turned their ideas into businesses. We got my first deal! Start your free trial at Shopify.com

SPEAKER_02: a Cosmic Queries grab bag, and only half the questions were from me. They were asking you questions, too. As it should be, Neil. The people, they need their Klepper. Give the people what they want. Come get your Klepper right up on StarTalk. Welcome to StarTalk, your place in the universe where science and pop culture collide. StarTalk begins right now. This is StarTalk. Cosmic Queries Edition, and this time, look who I have sitting right next to me, Jordan Klepper, oh my god! Can you believe it? Jordan, we love you, man. I like being loved. I go wherever I feel there is a warm energy that will bestow compliments on me, and I think I found it in this cluttered office. Right here at the Hayden Planetarium of the American Museum of Natural History. Half the time I put on Comedy Central, you're there. So it should be like,

SPEAKER_04: Klepper Central, they should rename the station. That's what I'm hoping. This is contractually, I have to do anything they want. I clean the bathrooms, I wash the windows, so it's just, it's Daily Show, Daily Show spin-offs, and then office reruns. That's what we are. You got it. Well, this is gonna be Cosmic Queries, grab bag, and you are my man in charge. Query master, which, fun fact, query is scientific speak for questions. If there's any confusion.

SPEAKER_02: If you're new to star talk, don't get lost on query. There's the people at home are tuning this off. They're like, what is this query? Already they're deep into it. It goes deep. So in Isaac Newton's publication of Optics, 1704, unusual because it was published in English. Helpful for those who only speak English. Yes, and if you don't always speak Latin, you're out. You're not in the moving frontier of science. I like that Isaac

Newton's already talking down to people. It's like, these English plebes, they're not gonna learn multiple languages. Let me give them the knowledge. So I think it was not until the second edition, which actually was in Latin, that he included a whole section in the back called Queries. And I just thought that was so elegant and so beautiful. And just so, well, because it was Latin, I think Queries is Latin. but that word has stuck with me ever since, because you read it, and he says things like, I wonder, but these are questions that are still left over after he wrote his book, okay? This stuff, the musings, right? So one of them is, I wonder if the stars in the night sky are just like the sun, except much, much farther away. Yeah! This is how he ends his book?

SPEAKER_04: It's beautiful, but it's also an unfinished book. If I got to the end of Lena Dunham's memoir, and she was like, I wonder what I did after Girls. Tell me Lena, that's why I bought your book. Come on, Isaac. No, he hadn't had the answer yet. I'm just saying, it was a whole set of them. It was just beautiful to read. These are the crumbs on his plate that he didn't know how to finish. That's beautiful, and you stole queries from Isaac Newton. Cosmic queries start talk give them to me the and these come in from our patrons our patreon supporters And they each pay five dollars a month for this which I think is a bargain actually well worth it Yeah, well worth it, and I've looked at something. There's some great questions here. Let's kick it off here. Let's do it Hello, dr. Tyson and Jordan Klepper. This is Cleo from Denver, Colorado And I have a question that both of you can answer from different angles You've interacted with people with wildly different worldviews Have you found that curiosity is something people choose or is it a trait that's surprisingly difficult to cultivate once we've settled into a worldview? Great question. Is curiosity something people choose or is it a trait that's surprisingly difficult to cultivate once we've settled into a worldview? That is a biggie. Curiosity, to me, is a beautiful trait. It may be one of my favorite traits. As a parent, I think a lot about curiosity. How old are you, kid? Five and a half. Oh my gosh. But I'm trying to teach my child to be curious, which is very easy for a kid, because I think children walk through space generally curious. Space and time. Yes, space and time. Oh my gosh, here we go. How many dimensions? Walk through space and time curious about the world around them and trying to find answers with that. And then I take that curiosity, I try to inject it into myself. I also then go out

into the world of politics. And I talk to a lot of people. who I get frustrated with their certainty and their lack of curiosity. Oh, so these are total strangers. I will say like, when I go to say like a MAGA rally, or it doesn't have to be even a political rally, but talking to people in the middle of America, when you interview people more often than not on camera, the camera, which also happens on social media, forces people to be much more certain in their views, and much less curious about the world around them, and only there to I think curiosity gets shut off. I see it getting shut off because people feel a fight or flight mentality to be more certain and right about the world view that they've concocted. So they're uncomfortable in uncertainty. Very much. Or even ambiguity. I think culturally, they're not rewarded for uncertainty. Online, you're only going to get retweeted if you have something.

SPEAKER_02: certain and hyperbolic. Right, if you say, I kind of think this is true, but I'm not sure, no one cares what you just said. There's no cultural reward for somebody who is curiously uncertain about the world. Ooh. You know, I don't think I have anything to add to that. Did I top you? Did I? You topped it and layered it and put icing on it. Did the student become the teacher? Did we watch it happen? I didn't go that far. No, let me try, let me see what I can squeeze into what you just said. So I, as an educator, I put it on me to ask, is there something I can dangle in front of them that will reignite their childhood curiosity in a way that might bring them back to reassess whatever it is they might be thinking? So you don't hit it blunt face on. You just say, hey, look at this shiny object, or look at this thing, or look at this idea. Well, that's interesting. And then you follow it through, and sometimes you can boomerang back on them and say, well, is that what I've been thinking all this time? But that takes effort on your part, the educator, because otherwise you're just stating what is true and hoping people come to you. But if they dig their heels in, they dig their heels in. And you need some act of persuasion more than just speaking what is true.

SPEAKER_04: I love that. I run into people who often ask me about changing people's minds, and think that it's done with just evidence or facts. That if I present the case, then the case will be understood, and more often than not, it is not. So instead of presenting the case, you present something

else that is auxiliary to the case, that's kind of interesting. They have less invested in whether it's true or not, and they learn about it, and then you work your way back to the center line. That's effort. That's effort. If we zoom out a little bit, if you're talking to somebody who has a set worldview, and call it an anti-scientific worldview. Earth is flat. Earth is flat. They're an Earth is flat person, they believe that. What are you dangling there in that conversation? Okay, so here we are on the East Coast, and middle of the winter. It's dark here by six o'clock. But wait a minute, there's a football game on in Los Angeles.

SPEAKER_02: We're in San Francisco. Let's turn it on. The sun is shining. The sun is shining on California right now. At the same time it's dark in New York. I wonder how you explain that. You're not jumping and saying Earth is not flat, you're just saying here's something that we can all see, that we all agree and understand that this is a phenomenon, and you can say one way to explain this is that. Earth is round. That's a way to explain it. Do you have another way to explain it? You just have that conversation that takes effort. It does. It's not sound-bitten. It's not a sound-bitey conversation. Well, and it speaks to your ability to create narrative. Oh, yes. Right? Yes. Yes. I hadn't thought about it that way, but that is true, because you're working a conversational way to get them back into the game of curiosity. And here's another one. If you lay down feet towards sunset, a perfectly clear horizon, so you gotta wait for that, okay? Feet towards the horizon, lay down on the ground, but part your feet so you can see the sunset. The instant the sun sets, immediately stand up, and the sun rises. We should do this all the time as a kid. Really? Yes. Well, I was a nerdy kid. Don't look at me like that. Shock. Oh, I'm shocked. You were a nerdy child. I was a nerdy kid. But in that instant, you come back, you were looking beyond the horizon that you previously had when you were laying down on the ground. That happens when Earth is round. That's amazing. So little things, and it's a fun little thing to do, and you engage people that way. So that's, it's effort. It is effort. But I will get back to your kid and say, All kids are curious. And the real challenge here is not how do you keep them curious? That's not necessary. How do you give them curiosity? That's not necessary. They're already curious. It's how to get out of their way. What happens? You got a five-year-old? That's the perfect age for them to go into the kitchen, open the cabinets, pull out the pots and pans, and start banging on them, okay?

And what are you gonna say? Stop banging. You're making a racket. You're making everything dirty. You just thwarted. an experiment in acoustics. I saved myself an argument with my upstairs neighbor, is what I did. Who cares about you? You didn't have kids to have a quiet household, or an orderly household, so everything they do creates disorder, and you want to keep a clean household, you should think differently. Everything they do is the, the mess they make is the consequence of an experiment they conducted. They don't think of it as an experiment, they're just exploring their environment. You don't think of an experiment, they're just destroying the environment. But as a scientist, they are cultivating their curiosity. Well, as a landlord, they are ruining my apartment. But I understand your point. Got another one for you. If we could travel faster than light, it would introduce paradoxes into the timeline of our lives. Some of them with no known way to resolve them. Which might mean it's not possible to travel faster than light at all. We don't yet know. If you like that fact, you can find 4,999 more in Lost in Space. 5,000 facts to help navigate the universe. Lost in Space is now available wherever you get your books. Next question is from Jack O'Melia Jordan. Oh, this is for me. Dude, it's my show, okay? From me, no big deal. No big deal. Dr. Tyson and Master Klepper. Oh, look at this. The respect of your Patreon members. Okay, so we all acquired titles over the years, and so Dr. Tyson's obvious, and Chuck Nice is here. He was bestowed the title Lord. That's very nice. We had a titled person from the US Space Force declare him to be Lord Nice, because he felt bad he didn't have a title. So ever since then, people have been acquiring titles. And so there you are, Master. Is that what he called you? Master Klepper. Okay. Do we think Jack O'Malia has the power to do so? Does he have the power to give me? What I do know is that in old England, Master was the title you gave to little children. I'm pretty sure. That feels about right.

SPEAKER_04: Childlike Klepper, Baby Klepper. There you go, Baby Klepper. Okay, what do you got? Dr. Tyson and Master Klepper. Greetings, this is Jack from Norwalk, Connecticut. Jack O'Melia says, Jordan, I've seen many of your fingers, the pulse segments, and they are brilliant. I love this guy. I know you edit it, so we only see the good stuff, but has there ever been time someone said something that set you back on your heels? Something completely unexpected that maybe we didn't see. Ooh. So first, let's verify. Do you edit your own? I'm usually in the edit, we have an

editor, we have editing bay, we take our pieces, we go in there, like all television, we edit it down, we interview usually about 12 to 15 people, edit it to like a four and a half minute piece. And about how many people make the final cut? I usually see a lot of them, probably eight to 10. And just to verify, just so people know, they all explicitly give you permission to put them on the air. They all sign an agreement. They understand the conversation. And so everybody you see has agreed to be on air. And you edit. Television is edited, and our bias is towards comedy, fighting hypocrisy, something that is interesting and compelling. But they're not super long interviews. They're usually anywhere between two to five minutes that we get little moments from. As far as something that stands out... I interviewed someone, I don't think this made an error. I interviewed somebody who was there on January 6th, who drove to Washington D.C. because they wanted a protest on January 6th. They went to the Capitol. They went inside the Capitol. They hit a cop, it was on camera. They were arrested. They then pled guilty to attacking a cop on January 6th. They served time in prison. And I talked to them after prison. After the pardon. The pardon, yes. And they were out. And they explained all of these steps, all these choices that they made to drive to DC, to go inside the Capitol. And just to be clear, you didn't pre-choose this person. this person happened to have all of that backstory. They had all this backstory, and this was a person I talked to who had written a book about it. So we found them, and they had a compelling story, and articulated all of these steps and choices that they made, and literally pleading guilty for the actions that they took, and I asked them who did January 6th, and they said, oh, it was the Deep State. And what kind of blew my mind is like, well, you did it. You made so many choices that day to be a part of the action. but you still can't wrap your head around the idea that the consequences you experienced were because of the choices you made. You were still being puppeted in some way. And I had an interview two days ago with somebody who was there on January 6th. It's sort of not to get caught up in the actions of January 6th, but it's such an interesting human phenomenon of people who both want to express power for the choices that they made for righteous reasons, and yet when pushed, want to still also hide behind the idea that they've been manipulated in ways that they couldn't have done the things that they did. And that happens often or often where you're like, people are like, there's so much hypocrisy out there. There's so many people living with two realities.

I think the deep state argument comes from the idea that none of that could have succeeded without a deep state enabling it.

SPEAKER_02: I think that was where that argument comes from. And I think there is a lot of like, well, why didn't they stop me when this would have happened? And more often than not, it is like these reactions you took and you can argue whether or not there were certain elements that were made easier. It kind of looked like the cops were trying to stop them. Was that just my, myopic view? Just my, from the camera angle I saw, it looked like the cops were trying to stop them.

SPEAKER_04: There's a lot of videos that seems like the cops put up a fight to stop people from getting inside. But you realize people find the media that tells the story they want. So that surprised you, that he could think that way, that deeply about it. Yeah, it was somebody who had built a bit of a reputation on not apologizing for their actions, but standing behind them as like, I was an actor in this because I saw something that was wrong and I wanted to make it right. And yet still had to.

SPEAKER_02: also be the hero and the victim in that story. So, the closest story I have to that, which doesn't come close at all, there was someone who was the husband of someone I was working with when we were filming Cosmos. An educated guy. And he doesn't think we went to the moon. And I said, why don't you think so? Well, there's no pictures of the landing site, you know, after we, plus we know we had to beat the Russians. So I said, what would convince you? that we landed on the moon. And he said, I think pictures of the landing site would do it, because we've been back to the moon with orbiting. So I said, okay, I found a website that had pictures of the landing site. And you can see the rover tracks and everything, it's all there. I gave him the link. The next day he comes back, Neil, I saw that, however, and by the way, it was a Japanese satellite that did this, JAXA is what it's called. And he said, I saw that, but that, Japanese satellite was in collaboration with NASA, and NASA's logo was on the page, and they're the government. So I can't trust it. And so I said, you asked me what would convince you, I delivered to you exactly what you requested, and you still refuse to believe it. At that point I said, we have nothing more to discuss. And I stopped the conversation. because it was clear he would just believe

it no matter what. Oh no, he said, I would believe it because he deeply respected me. He said, I believe it if you told me we landed on the moon, I will believe it. And I said, that's not how this works. It's not by my authority that we have landed on the moon. It has nothing to do with my proclamations. So he just went back to this conspiracy and I couldn't get him out of it. And I didn't have the patience to keep working it.

SPEAKER_04: I mean, that's the crux of so many of the divisions we have right now. There isn't that thing. If distrust is there from the get-go to sort of establish that trust. There's no opening. All right, we have a question from April Walsh. Hello, Sir Neil and Jordan. Sir Neil, this is April Walsh and my 16-year-old son, Aidan Free, coming from Windsor, Ontario, Canada. Our question is... When you look out as far as we can see to the CMB, is it the same distance away, for example, when you look east and west, north and south? Does it form a perfect circle that we are in the center of? I know there is no center of the universe, except in time. If it was, say, one million light years closer to the west, shouldn't we be able to see further than the CMB? Thanks in advance. And can I just add, what is the CMB? Stupid. I know.

SPEAKER_02: I just got a title a second ago and now I'm... Yeah, yeah, I rescind the title. It blew it. It blew it. No. CMB, I think I bank at CMB, right? They have very flexible hours for deposits. Wonderful hours for deposits. They don't charge a fee for me to take money out. So that is the Cosmic Microwave Background. Great. So I will hold you to that. The Cosmic, I'm writing it down in case you quiz me later. the cosmic microwave background, this is the remnants of the Big Bang, which were formed at a universe when we were like 3,000 degrees. Since then, because of the expansion of the universe, that light, that temperature has cooled to three degrees, three degrees absolute, so it's like three degrees Kelvin, not three degrees Fahrenheit, three degrees on the absolute scale. So what's happened is that heat, The intensity of that energy has been diluted over the expansion of the universe. So, that's a remnant of the Big Bang, and the Big Bang happened in the entire universe, so it looks like we are in the exact center of the CMB. The difference is, we are also moving in space. We are falling towards the Andromeda Galaxy. Falling towards it. Well, we're falling towards each other is the proper way, because there's no absolute

reference. where that's stationary and we're going towards it. And we're going to collide in five or six billion years. So put that on your calendar. I've got to put that on my Google cal. I've got a haircut that day. It's already prepaid. October 8th is when we collide, right? October 8th, five billion years from now. We'll cancel the haircut. So that collision will take several hundred million years to complete until it settles into a new galaxy. It'll probably settle into one double-sized galaxy. There'll be some... stars cast hither and yon from this collision, like a train wreck. But point is, we can detect our movement through space against the cosmic microwave background. Okay, so because we're falling towards Andromeda, that side, the side behind Andromeda is slightly warmer than the CMB that's behind us. Because we're moving towards it and that would give the impression to our detectors that it's slightly warmer in that direction And so we utilize that that that we utilize that to understand the speed with which we are falling for yes And so it's completely consistent right so we said can we see this movement in the cosmic background the answer is yes so because it's there and it's uniform and it's equidistant in every direction so so

SPEAKER_04: Deal with it. Alright, what else? Miranda Kagey. Hello, I'm Miranda from Austin. I want to know your thoughts on the mass of the Higgs particle. Is our universe fragile based on this and could a small change in one quantum after the Higgs field and change the physics we currently know? So I'm not qualified enough to give the answer I want to give to that question. But I will give a version of an answer and maybe

SPEAKER_02: I'll have to bring in one of them. I got people. I got people. Get your Higgs bros in here. I know Johnny Hadron. He's going to come in and collide some ideas off one another. We're going to get you the answer you need, Miranda. But Johnny Hadron, he's cousin to Jimmy Neutron. Those guys are fun drinkers, those two. So the Higgs boson. A boson is a kind of particle. Photon is a boson. Bosons are force. So they enable particles to interact with each other. Bosons as a category. The Higgs boson is in some circles called the God particle. Why? Because it creates a field and as you move through the field it hands you your mass. Now that's badass. That's pretty darn cool. That's totally badass if you're giving mass to other particles. And so I don't know the uncertainties in the mass of the Higgs boson itself and why that would matter to its power to give mass to

other particles. So I can't answer that. I need higher powers here. Don't make a face like that. That was like a disappointed face. You know, how about that maybe I could, if we don't have the answer, I can give you an anecdote about the Hadron Collider. What? Forgive me if I told this to you years and years ago. The Large Hadron Collider. Do you know, I was flown out 15 years ago to teach the physicists their improv.

SPEAKER_04: Is that right? Yes. Why? At the time, they hadn't found the God particle, because the Fermilab is outside Chicago, and there were physicists from Fermilabs who were big improv fans. Oh, because Chicago, that's a big comedy scene. It's a big comedy scene, and that's where I was from at the time, so I was learning improv, and sort of the birth of long-form improv is at a place called the Improv Olympic, and there was a physicist there who was inspired by the way comedians thought, and they were having trouble. over at the Hadron Collider. And so they flew myself, Sharna Halpern, who is the founder of the Improv Olympic, and my friend Seth Whiteberg to go over for a week to Geneva to teach the physicists how to listen and to work together. And so I spent five days teaching improv and theater workshops to a sea of physicists. How to make particle physicists funny? Yes. Like the German ones do. They all have to come together. They did a show at the end. And you know what it was? Terrible. It was a terrible show. Not super funny. But what I did learn, they were remarkable people, but the scientific mindset was so different than the improvisational one. So many people there had spent so much time defending their theories and their work that the idea of listening to somebody else and collaborating was an anathema to them. So we were doing very... basic improvisational exercises. Just like, listen to the end of this sentence before you start another sentence. And it was proving almost impossible for many of the people there. And after the few days, there were... breakthroughs on that level. And the point was, we have some of the smartest people on the planet here in this one space, and nobody is listening to one another. So let's let these comedians kind of help make a breakthrough. And then afterwards, they found the God particle. So is it? Oh, coincidence? I don't know. You said it. I mean, maybe. Maybe? So I have very deep respect for your profession. Thank you. And I consume.

SPEAKER_02: large quantities of it just because of the rhythms and the brilliance and the perceptions, perspectives, and so I, just so you know, I'm not worthy. But do you know about Alan Alda's project? I don't. Yeah, at Stony Brook, he may be a graduate of there, I'm not certain. He created a whole division of that school to teach acting to graduate students in the sciences so that they can know how to communicate with the public in ways. So they say, here's an idea, now deliver it in a convincing way and pretend like you're having fun. So the whole moniker is the Alan Alda Center for Communicating Science. You might have remembered he hosted a science program. We went around to labs and just talked to scientists and that's where he felt that rhythm. Comedy, if not acting, being funny, right? I mean, you're on stage, and some of the greatest actors we've ever had were formerly comedians. I mean, look at Tom Hanks, who's a comedian, and he's got, you know. The man's got awards, if that's what you're saying. He's got awards, that's what I'm saying. And he's winning a Toy Story, he has money as well, he's doing very well. I think you're articulating something that we were just.

SPEAKER_04: talking about is it's one thing to be able to understand the way the world works around us. It's another thing to be able to articulate that to somebody else who is skeptical of that. And in a way that makes them interested and want to come back for more. And that is Star Talk. See what I did there? Savvy. Oh man. Oh man, that's just easy. Keep it going. Here's what we got. This is from Steven Landau. Hello, Dr. Tyson and Mr. Klepper. From what we know about entropy, the fact that it always moves from lower to higher entropy, couldn't we surmise that the constant state of unrest we feel about government and politics is just another form of entropy? From my perspective, there has always been a political persona that has been triggering and considered the worst that has come. But that has always been one up by another that seems to become more hard-lined. Can this explain the fall of the Roman Empire, the American Revolution, World War II Germany, et cetera? Are these pivotal moments in history the example of entropy expanded to a rate that the system needs to reboot to low entropy? Wow. So this is deep. So we're talking about a new theory of political entropy.

SPEAKER_02: I'm game. Political entropy. Is that something natural? And if we build that into our governance, then maybe we're not so surprised by it. And that we would then navigate that as an inevitability. Yeah, to at least plan for the inevitability of where our politics is going to go. So here's why I don't agree. The Roman Empire was around for a long time. Where's your entropy there? No, it's building. There's no entry then it falls. Okay Then I was after the Greeks rose and fell. Okay, it just says things rise and fall You want entropy people talk about today? How turbulent the world is and how much unrest there is? You know, you can have a you know a van drive into protesters and kills a half a dozen people its headlines worldwide Yeah, it's bad However, I don't like lessening how bad one thing is by saying how much worse something else is. That's never a fun exercise. The internet doesn't like it either, Neil. Careful. Just contemplate this, and I've done the math. Between 1939 and 1945, 1,000 human beings were killed per hour. For every hour between 1939 and 1945 were killed in the service of the Second World War That were killed outright or died of disease related to being in a place where they couldn't treat it the desired of died of injuries thousand per hour and The fact that a half a dozen people could die from a van that plows into them and that make world headlines for a day national headlines for a week, local headlines for a month, and it was six people. You want to talk about entropy. There's nothing that rivals the Second World War, either before or since. And Europe has been mostly peaceful since 1945. Now, here's an interesting fact. As we withdraw support from NATO, saying, countries, you gotta put up or shut up, what's happening? Countries in Europe. are building their armies, which they didn't have to do before. If you don't have an army, who are you gonna declare war on? Who are you gonna fight? If you don't have an army, who are you fighting? You're just gonna build your economy and educate people and just have fun. Now you have an army because we're not footing your bill. And so I worry, and what do I know? I'm not a politician, I'm a scientist. But if you rearm Europe because we're stepping out of the support system, then I don't know, Europe has twice, has been down a road, nobody wants to repeat. So is this the buildup of entropy again, as armaments gather country to country? I don't know, but I think political science, which has the audacity to use the word science, in their. Oh wow, oh we struck a nerve. Blood drawn, okay. If politics were science, we would call it science. You can't tell me I'm not right there, okay? I'm a comedian, okay? I don't want to get in

this battle. Now you're backing away. I just occasionally read some political stories here and there. I don't know, I'm not getting in this fight. In politics, I think there should be a thread in their analysis. In political philosophy, there ought to be a thread of entropy.

SPEAKER_04: only happy when they're angry? Think about that. That's a great question. Are they only happy when they're angry? Let me be more precise. Are people only contented with their lives when they have something to complain about? I might get on board that anger is maybe the quickest way to give meaning to people. And in a society without, as a hard time giving meaning, or it's always hard to find meaning, anger is often the quickest way to mobilize it. Oh, oh, oh, oh!

SPEAKER_02: Science. That's science, Neil. That's political science. Join spinning political philosophy here on Star Talk. I love it. Were the cameras running for that? Did you get that? Retweet it, everybody. This is just, it's hot fire here on Star Talk. We got it. Yeah, this is what we do with your queries. Thank you, Isaac Newton. Well now, just to make sure it's real, can you repeat what you just said? No, I'm an improviser. I forget it all. That's what I'm saying.

SPEAKER_04: You just improvised a brilliant bit of political philosophy and now it'll never be recovered again. Just forgotten. Just let the people have it. It's time for me to think about something else. All right, we have another one here from Heather Walker. Salutations from Upstate New York. As a concerned parental unit, what is everyone's perspective on the less than stellar educational system and how do we rectify it short and long term?

SPEAKER_02: Hoping to see you at the museum Heather and Kevin Baez. Okay. One day I'll be writing a book on this, on education. Slow down, Neil. You're trying to get that 100, what do you get? Is it like the book it? You get like free of pizza if you write 100 books? You get a free pizza, free pizza. I get a little card punched. Every book, congratulations. So, let me comment specifically on science education. And then I'd like to comment on the arts, if I may. Please. So in science education, at least in the physical sciences, it's taught, well, here's a word, a vocabulary word, here's an experiment that was done, here's a fact, here's a formula, and then you learn it, and then you're done, and you go on to your next class. Science is a

means of querying nature and probing what is true and what is not, and ensuring that you're not fooled into thinking that something is true that is not, or that something is not true That is, that's what science is. And if that's not taught, which it isn't, then people come out thinking science is just a satchel of facts, and I memorize it for the test and move on, and nothing remains within them as a posture. They're not given a place to stand and to observe nature. So that when someone comes up to you and says, I have these crystals, let me rub them together and they'll heal your ailments. Had they been trained on how to probe nature? They would say, what is the mechanism by way they work? And what is the evidence in support of it? Can you show me, give me a demonstration now? Oh, did you know that crystals are the lowest energy state of the atoms and molecules that comprise it? So why are you saying there's energy coming out of the crystals? It has none. This is what you would do, okay? You have the power of inquiry. That is not taught. And that's gotta be taught. If we are to have a scientifically literate public, no, if we'd have a scientifically literate electorate, that's what matters. Because once there is no truth, objectively established truth by the methods and tools of science that are underpinning the foundations of civilization, once that objective truth goes away, there is no informed democracy. Preach. Preach.

SPEAKER_04: You are speaking, but let's expand that slightly more. Help me, give me some advice on this. I'm raising a five and a half year old right now who's going into school. It's too late. I blew it. They're on their way to school now. Just starting school. And so I'm thinking a lot about what type of education I'm wishing for my child, curating for my child, and I... I concur, I think discernment, understanding sources, understanding the information that you are getting, and I think about AI, I think about what is a toolkit that they're going to need 10 years from now? No, curiosity, that's the toolkit. If you're curious, you're always learning. Now, with your five and a half year old, has he found out that you don't know everything yet? No, I will not let that information get out. That information. There's a non-disclosure agreement with your wife? Yes, that information stays here. It does not get out. Do not tell my child. Do not tell my child. I think my child is figuring that out, yes. All right, so you should value not the answer, but the question. The question is the source of curiosity. The answer stops the curiosity, but the question.

SPEAKER_02: As it persists, it stokes curiosity. So if your kid asks you something you know the answer to, say, let's find out. What would be a way to find out? And get in the car and drive there? What color is the sun when it sets? Get in the car, go to a sunset, and observe, and think about it, and do things that feed the curiosity with your five and a half year old. and if your five and a half year old remains curious through those turbulent hormonal years, they'll come out the other side as a scientist, or as a minimum, scientifically literate, so they will never be exploited ever by someone who is ignorant of the laws of physics trying to sell you something to the contrary. How do I make my kid a venture capitalist? Do you have that? Because I think that's going to be much more lucrative. You want to secure retirement as they support you. Yeah, I can't help you there. Your things sound nice. That sounds nice as well. So now about arts education. I don't go out of my way to say this. So this is, I'm just letting you know, you've heard of STEAM. There's STEM, science, technology, engineering, and math. And there's STEAM, science, technology, engineering, arts, and math. Both sides of Congress know, except for aberrations, in the Trump administration, Republicans and Democrats historically have known the value of science as a force that stokes our economy. We have the world's largest economy that's been that way for like since any of us can remember. That's not an accident. It's what role science plays and how do you get the science discoveries into tech, into engineering, and how does engineering then become economy, right? Creative investments in science are the engines of tomorrow's economy. The arts people saw their budgets drop, and they say, let's piggyback the STEM movement, and that way whatever money goes to STEM, we'll get some crumbs that might spill off the table. I don't get in the way of that, but what I'd rather do is defend art on its own sake, for its own reasons. How do you do that? You say, oh, every city has some art sculpture somewhere in the park. modern office buildings, there's some sculpture you either don't understand, but it's there, it's there. You have the arts, there are people who sing to you in opera, okay? I don't know if operas are ever self-sustaining, I think they're not. They always require donations just to stay afloat. That's true with most of the arts, not the graphic arts, because we need those for movies and this sort of thing. But for fine arts, that needs support. Always, all the museums, there's like flow of philanthropic funds. Okay, so then you say, why do the art at all? Who needs it? You could imagine a country that has no art. And all the money

that would have otherwise been philanthropically given to art would go to building a factory and make something that they'll sell to you. Here's my question, Jordan. Were we Sir Jordan or Master Jordan? Master Jordan. Master Jordan. We can all imagine a country such as that with no art. But is that a country you want to live in? No. That's all I'm saying. It's very simple. I don't want to live in a country with no art, with no expression of human creativity, whether or not it can be commercially supported. I want to see the world through the lens of those who have a completely different view than I do. That's what art is, art at its best. And so, yeah, you take money away from art, you'll create a country that no one will want to live in, or it'll all be miserable, because yeah, we'll have money but no joy. We'll have resources but no perspective on ourselves, on society, on life.

SPEAKER_04: should be the future of education. Now that is an answer, Neil. Can I piggyback on that? Please. Because what I love... Please, you're not too heavy. Art significantly changed the trajectory of my life because it invited me to think about the world in a completely different way. It invited me to find emotions, live in uncertainty, understand... less about what was right and wrong, and more about who people are, how they feel. And I think art can articulate feelings in who we are in so many different ways, whether that's through fine arts, whether that's through comedy, whether that's through writing. Because we can feel and imagine, if that is denied us, then we are not living as complete a life as we possibly can, experiencing all that it is to be human.

SPEAKER_02: And one of my favorite paintings, in fact, is the very favorite painting of mine. It appears in a dozen different ways in this office. We should have a Weir's Waldo thing, okay? So, what's this called, Jordan? Well, that is Starry Night by Vincent van Gogh. Very nice, except you left out the. It's a the Starry Night? Yes. Idiot. Dammit. Dammit. Okay, so here's my point. I'm an astrophysicist. I'm telling you right now, the sky has never looked like this. Not even for him on the night he painted this. I know this. In fact, separately behind my desk, I have a printout of the actual stars, moon, and planets in the sky when Van Gogh painted Starry Night. Here it is. So this is the actual night sky. And how do I know the night he painted it? How do I know? Because there's a crescent moon and the planet Venus. That's Venus down there, the bright one. That configuration at that angle to

each other, at that elevation above the horizon, tells me he painted this somewhere around 4.30, 5.30 in the morning, near the June solstice in the year 1889. In Remy de Provencial. Did I pronounce that right? Rémi de Provençal. Don't turn to me for this. I do not have your answer there, yes. Anyhow, point is, this is not what he saw, but looking at that night sky, this is what he felt. I want to experience the world the way artists feel it, thereby giving me sight lines to all the ways of being human, enriching my own life with your creativity.

SPEAKER_04: That's for me the future of education. That's worth your pet's patreon money right there. Am I right? That's beautiful invest in the arts and the sciences make them come together I just cuz that is life and the pursuit of happiness Somewhere I read that That's the political sciences, your second favorite science. Political philosophy, let's call it that, yes. Okay. And also, I will say, if you don't have the arts that can help you articulate the sciences, then you're just alone in your room without... Like, I came up with something great, but nobody will listen to me. You should have taken an art class. Let's bring in an improviser. This from Mark Hart. Hello, Dr. Tyson and Mr. Clapper. Jordan, if I may, this can be for all of us, if I may, how do you keep from becoming depressed and hopeless? What drugs are you on is the question. Well, I'm on allopurinol for my gout, so. What are you, King Henry VIII? Let me tell you, I live a life of privilege. If you have gout right now, you are living a good life. Those purines, the steaks. Huh? The yeasty IPAs. The turkey leg, yeah, okay. Oh, this is, yeah, I wear a gout like Gucci slip-ons, just as proof of my success. How do I keep from being depressed and hopeless? This is a question I think we both could, I like to unplug, is one. I think the quickest way for depression, for me specifically, is to get lost in the fake reality that is online discourse. And so wherever I can unplug. I will, you know, we're working on the same themes here. I went to MoMA last week, where Starry Night is. The Museum of Modern Art. The Museum of Modern Art. I took my child there. We go usually a couple times a month. Nice. We love to draw the pictures. We saw the Starry Night. So I have the original behind my desk, so this is a copy of mine. Oh, that is the original. Behind my desk, it's a copy. I was going to say, yes, it's beautiful. It really is. And you deserve it. But don't tell anybody. No, I won't. No, you're fine. As long as nobody listens to the podcast, you're totally good. I like to dip my head into abstract. You go to a

modern art museum and look at a Jackson Pollock, and nobody's telling you what to think. It's just expressing a feeling that can't be articulated in words. Jackson Pollock, the splattered paint genre. The splattered paint, or Rothko, the big loud paint. All of those to me are sort of a reset that helped me step away from the depressing things. And then the cliches of family. Stepping in and paying attention to a little child who's five and a half who needs you who needs you. Yeah It gives you meaning it gives you purpose and it gives you perspective that they care a lot less about what was on the news that morning It'll hear anything. They don't care at all. Right? Yeah Well, I mean my little ones a kind of a Bernie bro, but you know the five and a half They'll grow out of it. But so I get a lot of progressive politics But outside of that we can move into Legos and you could find a bit of joy

SPEAKER_02: I don't know that I can add to that. That was beautiful. And whiskey, whiskey helps a lot. So I'm generally a happy guy. It's not an act. I'm generally, but I'm an optimistic realist. I'm not going to delude myself into thinking something is better than it is, because then solutions miss their target if you don't know exactly what's happening and why. I always carry the hope, and I don't like the word hope, because hope and prayer are what you have when you confess that you are not in control of the situation. So I don't want to say I have hope, so let me say I have confidence that with some effort and with some time, it will change for the better. As the history of civilization has, I am today not owned by someone on their plantation, okay? I know that's a low bar. It's a very low bar, but it's a good thing. Okay, yeah, we can be grateful for many things. Okay, you know, pick your oppressed group. You were not better off in the past than you are today, statistically, if you just look at overall, and yes, there are fits and starts to that arc. I look at where we were versus where we are and I say that's progress, let me keep at that. I grew up at a time where I was not welcomed for my scientific curiosity, urged to do athletics because in people's expectations my skin color were only imagined in that role, not as an astrophysicist. I can speak for hours about how many times I've been stopped by the police, by security guards, assuming I'm a criminal, and I could go on and on. However, my father's stories were way worse. And his father's stories were way worse than his. So I try to step back, get a longer measure of that arc, and say things are better. And Jordan, I've written about this. in one of the books that I had hole punched. You're gonna get that

pizza sooner than later. May I? Please. May I? Starry Messenger, Cosmic Perspectives on Civilization. This came out a few years ago. I've read it twice. Did you? Yeah, I loved it. Why'd you read it twice? Well, because I was bored by it the first time, so the second time I was like, I must have missed something. Okay, one last thought experiment. Imagine you're offered the opportunity to enter a time machine and go back to any previous moment in human history. Where and when would you choose? The white, cisgender, heterosexual male. Check. Is there one in the room? The heterosexual male has his pick of the timeline. This traveler. to the past will be welcomed everywhere and every when. I invented that word for that sentence. Every when? Yeah, it's not in this dictionary. You need to get an editor for these. No, stop. Leave me with my new words. Everywhere and every when. If that's not you, then you'd better think hard about the time and place you'd like to visit. Are you female? a person of color, disabled, queer, or any combination of these? When was it better for you? A thousand years ago? 500 years ago? 100? 50? 10? Five? For me, I'm good with the present. I'd rather not be declined a taxi ride, overlooked for job opportunities, denied a bank loan, or red line from my choice of housing. With childhood ambitions of being a scientist, I don't want to be someone's servant, nor do I fancy being purchased and owned by another human being who thinks I'm not entirely human. Come to think of it, I'd rather visit the future. as I presume the preternaturally progressive Unitarian minister Theodore Parker would too, as he wrote in 1853. I do not pretend to understand the moral universe. The arc is a long one. My eye reaches but little ways. I cannot calculate the curve and complete the figure by the experience of sight. I can divine it by conscience. And from what I can see, I am sure it bends towards justice. That's the origin of the comment, the arc bends towards justice. So, how do I stay happy, positive, with hope? I look at how much worse it was in the past, which means there has been progress. If not immediately today, or in the last month, over my life and over the life of everyone who has preceded it.

SPEAKER_05: That's how and yes the kids part and the raising the kids Yeah, I said bourbon and then Neil's like get my book, please Bourbon I said bourbon and a sip of wine every now

SPEAKER_04: Time for a few more. A few more, let's see what we got. This is Joe M. Hello, Dr. Tyson and Jordan. Joe M. from Denver, Colorado here. That's a second from Denver? We got Denver folks here, okay. Shout out to Denver. I've long said that it isn't my responsibility to convince people they are wrong. I'm not their parent or teacher. However, it seems that people have fallen so deeply into the trap of only believing the evidence that supports what they already think that they may never see anything any other way. It almost feels that today what folks believe feels more like religion and faith than anything else. Belief despite evidence. When having conversations with people like this, how can we spark curiosity so that they can start to look for and take in evidence themselves? With boundless curiosity, Joe. One of my favorite things not taught in school, we need an entire class on psychological bias. Confirmation bias. That's the most...

SPEAKER_02: pernicious. Is that a right, did I use the right word? That seems close. It feels close to the word. The most, pernicious, that's the word. I pay attention to what supports what I already want to be true, and I barely notice conflicting evidence. And therefore I build a case that supports how I think the world should be rather than how the world is. That's. confirmation bias. There's a wiki page on all the kind of biases that we have. There's other ones. You buy a brand new kind of car, and all of a sudden, everybody has that car. You ever notice that? That's a bias. You have an awareness that changes your awareness of other things. That's a bias. In science, we are trained to reduce our biases to as little as they can possibly be, recognizing that we're human, and we're going to have biases no matter what. So as a scientist, I say, I think I have no biases here. Could you check this? That's peer review. You're a peer. You check it. You have a different configuration of biases. You might be able to spot mine that's not yours. And peer review is a way to keep bias in check, even though we are nonetheless all trained as scientists to reduce the biases as much as possible. The public, with no such background, all hell breaks loose. The biases are running rampant. The solution here is go back into the school system and teach an entire class on bias. This whole reading, writing, arithmetic, we gotta up our game there into the 21st century. How to spot something that is not true, in spite of your bias. Leonardo! There's only one Leonardo. Well, yeah, I believe he was the one who ate pizza, which one had a Nutella, or

no, Michelangelo ate the pizza. Leonardo was the one with the... Pre-dating the Teenage Mutant Ninja Turtle. Tell me more. Yes. There was Leonardo of Vinci. He says, and I quote, the greatest deception men suffer is from their own opinions. He knew it. 500 years ago, this man knew it, okay? So that would be an antidote to all of those problems. Well, I stumble upon this all the time in my work. The only time I see movement was somebody who feels

SPEAKER_04: in a point of position, and they're strong in their opinions in a way that they don't want to change, is admitting your own failure, pointing out your own flaws and uncertainties. How quick are you to do that? Yes, well, I am not quick to do that, but I think within my families and within people who I find, again, off-camera this week, I had the one productive conversation that was actually moving towards some sort of understanding and or truth came, off camera when I was able to show and articulate the things that I was unsure of, the things that I didn't know. And then that person then met me with love. You were opening up your soul. Well, I think to me, just as a human being who came up in the comedy scene, all of my closest friends, I've watched fail over and over again. It's what improv comedy is, it's what life is. And to me, trying and failing and trying again is human.

SPEAKER_02: who's truly successful hasn't failed. Exactly. And I think people pretending as if they are geniuses and or are perfect in their thoughts and their ways to me is an insincere projection of who you are. I would use the word delusion. Delusion, exactly. It's a self-delusion. It's a self-delusion. Yeah, yeah. Someone once asked me, when was the last time I was wrong? And I thought about it and I said, let me do a whole show on that. So we have what we call an explainer. and Chuck was with me here, Chuck Nice, and I just said, I'm wrong all the time, okay, and there are levels of wrong, there's like, I'll be home by six, no, I'm home at 6.30, I was wrong, okay, that's probably not where they were headed with that question, but you can work your way up the ladder, and the point is I don't have emotion invested in what I think, it's proportional to data and evidence, so if the evidence shifts, I'm shifting with the evidence. Immediately. Okay, if I don't have emotions invested in it, I got nothing, I'm not shackled back to with heels dug in saying, it's gotta be this way, I'm not changing my mind. I change my

mind all the time. Bring on the evidence and I'm there. That's all it is. So the question almost doesn't mean anything to me. That articulates something that isn't talked enough about. It's the emotions that are invested into that.

SPEAKER_04: that also in a political perspective, identity then gets built on top of those emotions. And then you can't mess with the identity, because it's built, and even if it's a house of cards, that's your identity. That's who you are. That's who you've been projecting to your family members, the people you love for 10 years, and now somebody gives you a piece of information, you have new data points, and you're like, I have too much built on this idea. Jordan, we got 10 minutes left, four questions.

SPEAKER_02: Let's close the lightning round. Okay, go. Okay, Danita Buchheit says, hi Neil Jordan, Danita from Utah. What's with this space reflecting mirror? Is this crazy? Are we doomed? So one of the latest ideas was to put a huge reflecting mirror, it's actually a Mylar surface. Mylar is very reflective, very thin, very light. So it doesn't take much energy to get it up. You deploy it, it's like. 30 yards across, I mean it's huge. And it would position it in such a way that it would reflect sunlight back down to like a five kilometer area on Earth. For what reason? I couldn't wait for you to ask. It's cool. It's like why? Okay. I think we have an object up in space that does a lot of sunlight on a certain, on a lot of Earth. Called the sun. Yeah. Seems pretty effective. So, and I might want to sleep every now and then at night. So you wouldn't do this in the daytime, there's no point, because like you said, the sun is there. Nor could you do this if it's cloudy, because it would get blocked by the clouds. You would do it when it's clear on the ground, and it would be a way to get a few extra hours of sunlight beamed down onto your solar panels. Now, when I say a few extra hours, it's high up enough so that when the sun sets for you, it hasn't set for it.

SPEAKER_04: Further evidence, Earth is round, by the way. So it can keep beaming well deep into twilight, but in the middle of the night, it doesn't see the sun, it's in Earth's shadow, so that wouldn't work. And it's getting like, you said like a five mile radius? I said five kilometers. Five kilometers, I can't even hear kilometers. It's just like, and I was already, I was like, just make it miles, just don't get into it. Because you're American. American, that's right.

SPEAKER_02: That's right, Isaac Newton, write in English for me, all right? Plain English. And he's English. He would've used miles and Fahrenheit. So he's your guy. That's my guy. So yeah, it would create a bright source of light in the sky. Now, of course, it's beaming it to one location or another. It's not lighting up the whole sky. But the plans are to have thousands of these in the sky. That's the plan. And gone is the night sky. The privilege that I grew up at a time, even though I was in the city, you go out to the country, the night skies are a majestic canopy of the cosmos looking back down upon you. And now it's a canopy of satellites. Point is, back then, in twilight, I would catch one or two orbiting satellites. And they're fun because they're very different from an airplane. and they're moving fast. They say, check this out, it's an orbiting satellite. Oh my God, we have satellites up there. Two, three weeks ago, satellites were just going back and forth. It wasn't even interesting. It wasn't even special. Most of them were SpaceX Starlink satellites, which are now rising through 12,000 in the night sky. Yeah, SpaceX is like 80, 90, 85% of all satellites in the sky have been launched by SpaceX right now. So now it's just, criss-crossing satellites, all I can say is this ship has sailed, and I don't know if there's any coming back from it. The night sky is no longer sacred, as it was to the ancients, as it was to anyone who didn't have a Netflix account at night, who would not otherwise be distracted at night, to look up and bask.

SPEAKER_04: Just a few minutes ago, you were talking about how much better it is now than everything in the past. The arc, what happened to the whole arc of morality? You got a book about this, Neil, huh? Got a quote you want to read about this? We'll get spaceships to go to the moon, the backside of the moon, and we'll have observing trips there. Which would be way better than anything we ever saw on Earth. Would I get an invite to that party? Yes. I'm holding you to it. Okay. Real quick, we run out of time. Real quick. All right, this is Natasha from New Mexico. Hello, Dr. Tyson, Star Talk team, and Jordan Klepper. As an incoming MD three more years, my field is currently at the peak of unfortunate theories about vaccines. How do people become anti-vax? Why? Is there any good way to help people understand their importance without losing herd immunity? Do human beings have to FAFO to learn? Saying this as a thick-skulled woman, please give me hope.

SPEAKER_02: Natasha from Mexico, New Mexico. I was executive producer on a documentary called Shot in the Arm, which is an exploration of vaccine hesitancy in the country and in the world. Congratulations. And ways to navigate where that hesitancy is coming from and how to overcome it. Shot in the arm. Google it. Are you gonna give us a taste? No, because I spent the whole time being executive producer and I was science advisor because the filmmaker himself Scott Hamilton Kennedy He has an active Instagram page Scott Hamilton Kennedy is a filmmaker see he it's his film But it has science themes and he wanted to make sure that they were communicated accurately or as effectively as possible and Because he's a storyteller and so I came in and helped shape many of those stories. So it's a sympathetic look at those people who were duped by anti-vaxxers. And so I recommend it. It might be streamable on PBS, but find it out there. Yes. Mm-hmm. This is from Khalid Alves. Hi, y'all, this is Khalid from Brazil. I listen to StarTalk Podcast every day, and I'd like to know if Mr. Klepper ever got scared doing his job. Ooh, I like that. All the time. Wait, wait, wait. Just to be clear, when we see you on TV, you're just talking to a camera. So how scary could a camera be relative to live people who could throw things at you? I mean, this camera steals your soul and it makes you a fame whore, so you become a terrible person. So that's one thing to be afraid of. And I've seen comedians who went all camera for many years, they came back to stand up and they lost some mojo.

SPEAKER_04: They think they're funny just because they think they're funny, but they had no evidence that they were funny in front of a live audience. You guys, without a live audience, can get rusty. You can very easily get rusty. And the great thing about a live audience is, I mean, you get real-time responses as to whether or not you're effective or not. In the moment. It's like, okay, that didn't work. Yeah, I mean, I've done... I go to places like rallies where things can get dicey, but I have four security guards. Political rallies. Political rallies, and I was there on January 6th. I've been at places where it gets nasty, but on a performance stage in front of an audience, it's probably the most nervous and scared you get because you're confronted with the things that you think are interesting, compelling, and or funny, and then an audience of a thousand or so people who tell you...

SPEAKER_02: No! We don't agree. Right. By their silence, they don't agree. They don't, yeah. Yeah, there's nothing scarier than a blow to your ego. I think that my physical harm is something I'm fairly okay with. But the emotional harm, you're still sensitive. And what does that do to the remainder of the show if you don't get them up until some point? And you gotta still deliver the show. You still, you have, that's...

SPEAKER_04: That's the work. That's when you dig in and like, all right, I have a show to do. What you get better at. I was doing new material a few weeks ago. You get better at. at loving the bomb and loving what is bad about it. I was doing some new material about things that I saw online that I thought everybody was having an experience with, and as soon as I started talking about it, I realized nobody was having that experience. The algorithm had lied to me, and I had crafted a bunch of commentary about the things that I was seeing, and nobody was seeing it. And it felt terrible, but also amazing in that, like, I've gotten to the point where, like, well, this is...

SPEAKER_02: very interesting. This is not, I wish this were funny, but now I'm gonna explore that I'm having experience that's completely alien to yours. I'm gonna talk to the people in the audience. I'm gonna try to find out more about it. And you love that. You made lemonade out of lemons. Yes. So, three weeks ago, I deleted my entire search history on YouTube. Really? Yes. And then all of a sudden, recommendations came up and said, was that what other people were looking at? Yeah. Yeah. It's buried in there, but you can do it. And it's a fresh start. I recommend it. We are missing out on serendipity. That's a beautiful thing. And let me give my version of what you just described. When I'm on stage giving a public talk, I like saying funny things. I like making the audience laugh. I deeply respect what you guys do. I absorb as much as I can. But if a joke doesn't land or they're not laughing, I'm still teaching them something about science. So you don't have that luxury, okay? You don't say, okay, I'm not funny now. I will be educator for 10 minutes. You don't have that luxury. You don't know my material. If I bomb, I'm like, all right, we're going into quarks. We're just gonna talk quarks. They're gonna get something out of this. Right, they will get something out of it even if nothing I say is funny. They went to a lecture. So that's how I survived that.

SPEAKER_04: I think that's smart. Work on a couple different levels. That's a smart move. All right. Last one. Last question. This is Myann from Phoenix, Arizona, just a few kilometers from the sun. If you'll indulge me, I have a thought experiment for you. Love them. If a roughly five light year wide bubble around the sun were suddenly displaced to another region of the universe, carrying our solar system and the light already traveling through that space with it. Oh, I love it. our familiar night sky might appear unchanged for years before the old photons finally ran out. When the new sky arrived, what would the transition actually look like from Earth? And what would scientists detect first that proved we were no longer in the same part of the cosmos? I'll take this one. That's a fascinating. That's a beautiful question. That's fascinating. That's like sci-fi fascinating. Yeah.

SPEAKER_02: So that means we have five years of the old sky. Oh, look at the Big Dipper, and look at this. And then in an instant, the entire sky would change. In an instant. And all the constellations would be different. There'd be no Big Dipper, no Cancer, no Leo. The astrology would have to reinvent itself, because... we're in a different part of the galaxy, and all those stars were only visible from where we are in our part of the galaxy. By the way, in 100,000 years, since we're in a rotating galaxy, all the stars that you see tonight would be in a different position anyway, but you'd sort of watch them drift over the 100,000 years. So constellations today would be very different from 100,000 years from now, and if it's a five light year bubble, they will change instantly, and that would be fun to watch. A more devastating would be if a giant plucked the sun from the middle of the solar system, plucked, okay? What would happen, okay? Someone actually asked that of Merlin, which was my pen name, for 15 years. I wrote a column and I wrote a book. This is one of the hole punches. This pizza, it is on its way. And Merlin's tour of the universe, this came out a couple years ago, one of the questions was? If the sun were to disappear all of a sudden, how much time would we have to think about it? And my brother who's an artist, Drew, I don't know if you can see. And it shows some cosmic hand plucking the sun. So how much time would we have to think about it? First of all, you wouldn't know what happened until that information reached you. And that takes eight minutes and 20 seconds at the speed of light. So not only does light take eight minutes and 20 seconds to reach you from the sun, the absence of the light takes eight minutes and 20 seconds to reach

you. That trailing edge of light from someone having removed the sun. So we will sit here, orbit the sun, rotate, bask in the energy of the sun on the beach, then plunge into darkness and fly off at a tangent and lost forever to interstellar space. Have a nice day No, no, but it's just fun to think about because the question was how long would you have to think about it? You wouldn't know what happened until it was too late for you to do anything about it because and that's the eight minutes in 20 seconds That's 500 seconds. So you can only interact with the universe with the information Coming to you at any given moment. If you got a heads up, what would you do for those eight minutes? Who would give me the heads up the Giants? Oh, you're like, yeah. Let's say you got intel with the giant. He's like, I'm not gonna save you, but. I'm gonna do it at this time. I'm plucking the sun, so you got eight minutes, good luck. Okay, I would... Everyone should just kiss their ass goodbye. I think that's the... What do you do? Move near volcanoes, because that would be your only heat source. Oh, okay. After the sun is gone. That Mount Etna down there in Sicily, that's beautiful anyway. Is it still erupting, Mount Etna? It was recently. Oh yeah, okay, yeah, just move inside the crater. Okay, go into the crater. Try not to inhale the poisonous gases, and you'll be fine for a while. Then I give you what, a little bit? Well, yeah, so all the plants will die, and then all the animals that eat the plants will die, and there'll be a wave of extinction percolating across the tree of life. And so, depending on how much canned food you have, so for example, I have, so what is this? That looks like rice and beans. It looks like chicken soup. It's soup. Okay, look what kind of soup it is. Wallet? It's primordial soup. Good, I've been looking for that. You need this in situations like that. You can heat it up on the lava and jumpstart civilization on Earth after we all die. So do your shopping and move to Sicily. Jordan, that's all the time we have. Thanks for coming in and doing this. I know you're a busy guy, and there's always a rally out there where we see you querying.

SPEAKER_04: participants. It's so much more fun and enlightening to query with you every now and then. Well, thank you. It's a nice rejuvenation and a connection to the world as it exists. To rational reality. Thank you for that. To the objective truths of the cosmos. It's nice to touch grass with that every now and then. So are you appearing anywhere? Where do we find

you? You're going to find me at The Daily Show a lot of the time, so if you got a TV, check me out there. I also travel the country and I do my stand-up.

SPEAKER_02: so check out officialjordanklepper.com and I might be coming to your town and hanging out and telling some jokes. So official is part of the web address? Official Jordan Klepper, yeah. Don't go to Jordanklepper.com. I don't know who has it. There's some fake people. Don't trust him. Don't trust the fake Jordan Klepper. Don't trust the fake Jordan. So we can track your comedy tour on that. Love it. This has been Star Talk Cosmic Queries, Jordan Klepper edition. Oh, I like saying that. Until next time, as always, I bid you. Keep looking up.